

MTH 446: Lecture # 14

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September 28, 2026

Lecture Span

1. Properties of complex derivatives

Complex Differentiability

Remark, if $f \in H(D)$ and $f = u + iv$, then u, v are infinitely differentiable.

Definition 0.1. u is said to be harmonic conjugate of u if

$$u_x = v_y \quad (1)$$

$$u_y = -v_x \quad (2)$$

In D . We abbreviate harmonic conjugate as h.c.

In other words, if $u, v \in H(D)$, then v is a h.c. of u in D iff $f = u + iv \in H(D)$. How to find a harmonic conjugate?

Example

Let $u = 2x - x^3 + 3xy^2$. Find v , the harmonic conjugate of u . We can verify that u is harmonic. Use the C.R. equations to obtain that

$$v_y = 2 - 3x^2 + 3y^2 \iff v = \int dy(2 - 3x^2 + 3y^2) \quad (3)$$

Then, we obtain

$$v = 2y - 3x^2y + y^3 + C(x) \quad (4)$$

Then

$$v_x = -6xy + C'(x) = -u_y = -6xy \quad (5)$$

Therefore, $C'(x) = 0 \implies C(x) = \text{const.}$ Then,

$$v = 2y - 3x^2y + y^3 + C \quad (6)$$

Example

If v is h.c. of u and $\tilde{v}$ is h.c. of u , then $\tilde{v} = v + C$ for some const. C .

Example

Suppose u, v are h.c.'s of each other, then u, v are constant in D .

Proof. This means that

$$u_x = v_y \quad (7)$$

$$u_y = -v_x \quad (8)$$

And vice versa. Then, we arrive that

$$u_x = -u_x \iff u_x = 0, \in D \quad (9)$$

We obtain similar equations for v_x, v_y, u_y . Therefore, $\nabla u \equiv \text{const.}$ and $\nabla v \equiv \text{const.}$ $\square$

Theorem 0.2. *Reflection Principle: Suppose $f \in H(D)$, and contains segment of the x axis and is symmetric to the axis (lower half in D is reflection of the upper half w.r.t the x axis). Then,*

$$\overline{f(z)} = f(\bar{z}) \quad (10)$$

if and only if $f(x)$ is real for every x on that segment.

Example

Suppose P_n is a n -th degree polynomial with real coefficients. Then, one can prove that

$$P_n(\bar{z}) = \overline{P_n(z)} \quad (11)$$

Definition 0.3. *The exponential of z is*

$$\exp(z) = \exp(x) \exp(iy) \iff \exp(x) (\cos(y) + i \sin(y)) \quad (12)$$

This is the exponential function.

Properties

1. $\exp(z_1 + z_2) = \exp(z_1) \exp(z_2)$
2. $\forall z \in \mathbb{C}, |\exp(z)| = e^x > 0$
3. $\forall z \in \mathbb{C}, e^{-z} = 1/e^z$.
4. $e^{z_1 - z_2} = e^{z_1}/e^{z_2}$
5. $\exp(z)$ is a periodic function with period $p = 2\pi i$ for all $z \in \mathbb{C}$. Note, the period is a complex number. *Proof:* $\exp(x) \exp(y + 2\pi) = e^x (\cos(y + 2\pi) + i \sin(2\pi))$. Therefore, $u, v \in C^\infty(\mathbb{C})$.